Two View Epipolar Geometry

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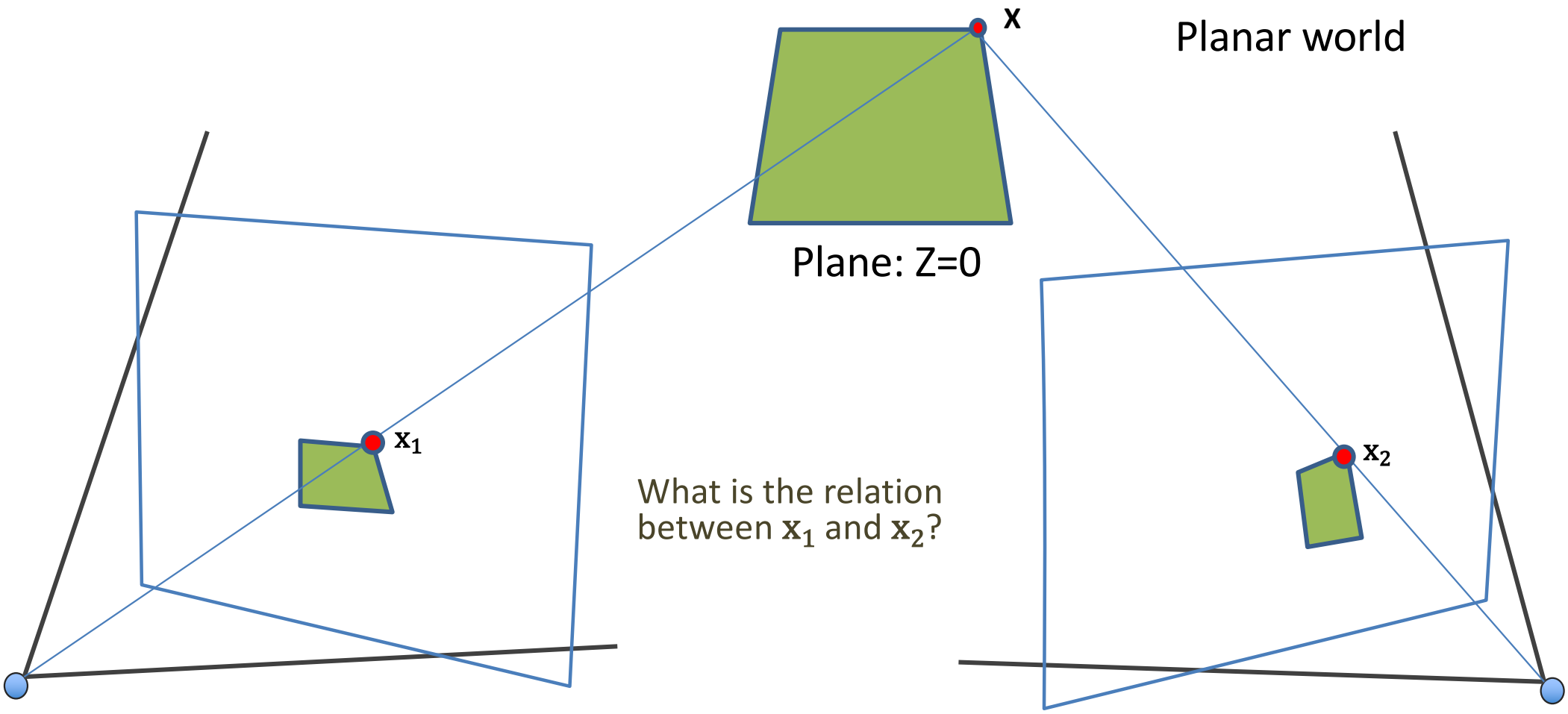
Image courtesy: https://www.etsy.com/in-en/listing/1607436850/third-eye-activation-spiritual-awakening

Lecture Plan

- Two View Geometry
- Epipolar Geometry

Case 1: Planar Geometry



Case 1: Planar Geometry

- Projection equation of points on a plane:

$$\mathbf{x} = \mathbf{K}\begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix}\mathbf{X} = \mathbf{K}\begin{bmatrix} \mathbf{r_1} & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{t} \end{bmatrix}\begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix} = \mathbf{K}\begin{bmatrix} \mathbf{r_1} & \mathbf{r}_2 & \mathbf{t} \end{bmatrix}\begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} = \mathbf{HX},$$

where H is a 3 × 3 non-singular matrix

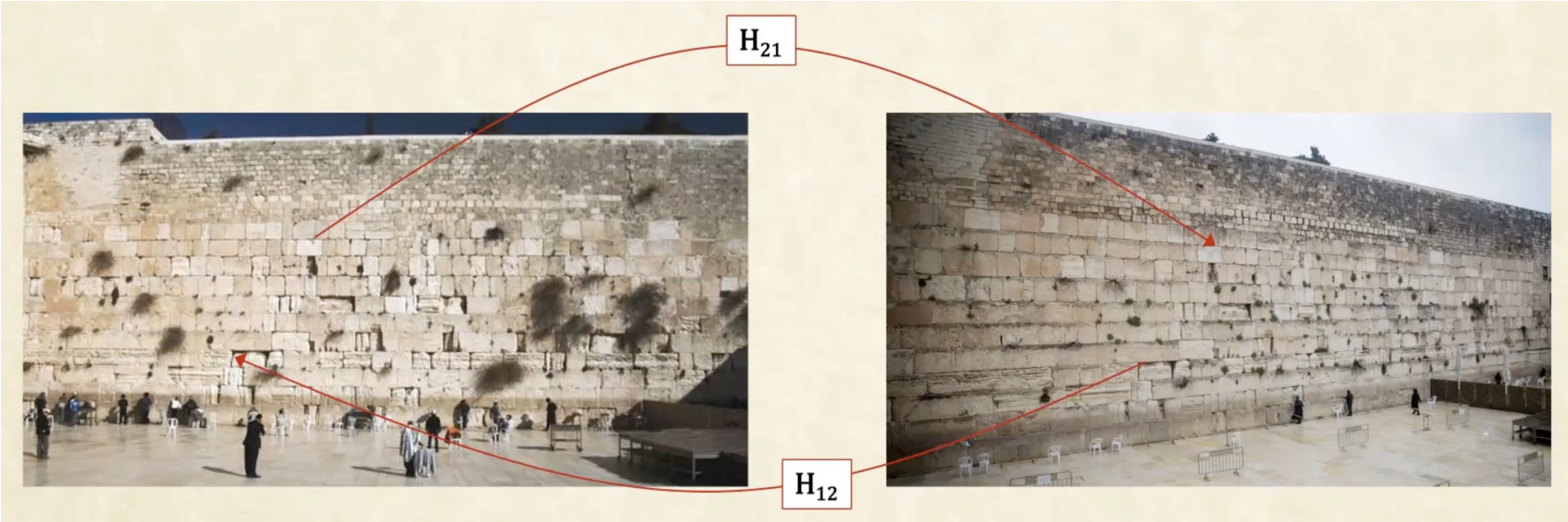
- Now if we consider two different views of the same world point, we get:

$$\mathbf{x}_1 = \mathbf{H}_1\mathbf{X}; \quad \mathbf{x}_2 = \mathbf{H}_2\mathbf{X} = \mathbf{H}_2\mathbf{H}_1^{-1}\mathbf{x}_1 = \mathbf{H}_{21}\mathbf{x}_1.$$

$$\boxed{\mathbf{x}_1 = \mathbf{H}_{12}\mathbf{x}_2; \quad \mathbf{x}_2 = \mathbf{H}_{21}\mathbf{x}_1}$$

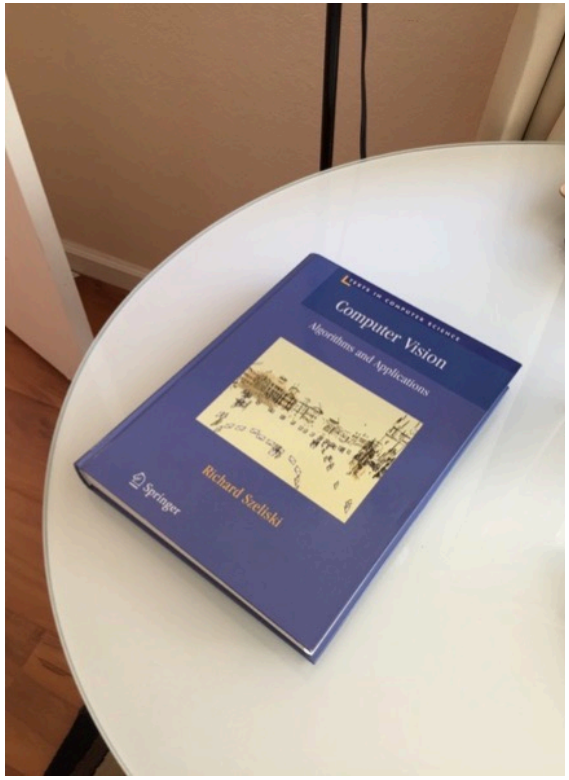
Case 1: Planar Geometry

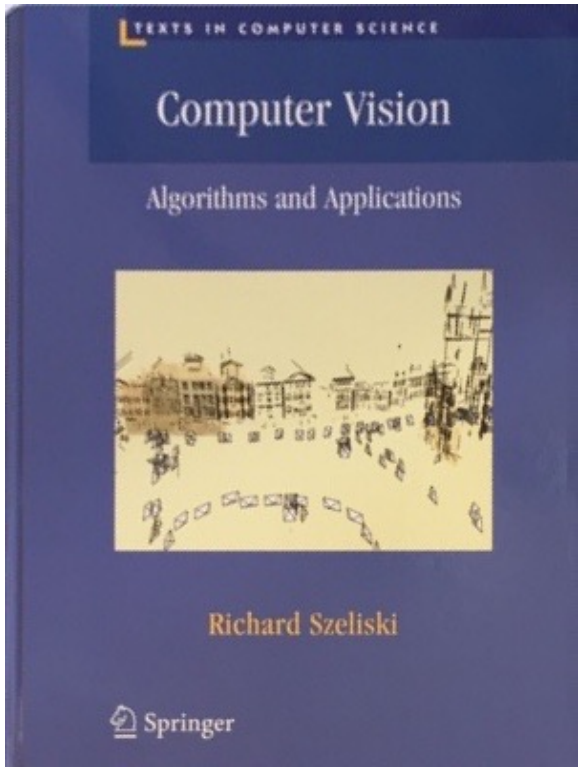
- Given two images of a planar world, every pixel in an image can be computed from the other using homography matrix **H.**



Applications of Planar Homography

- Correcting Perspective Distortion



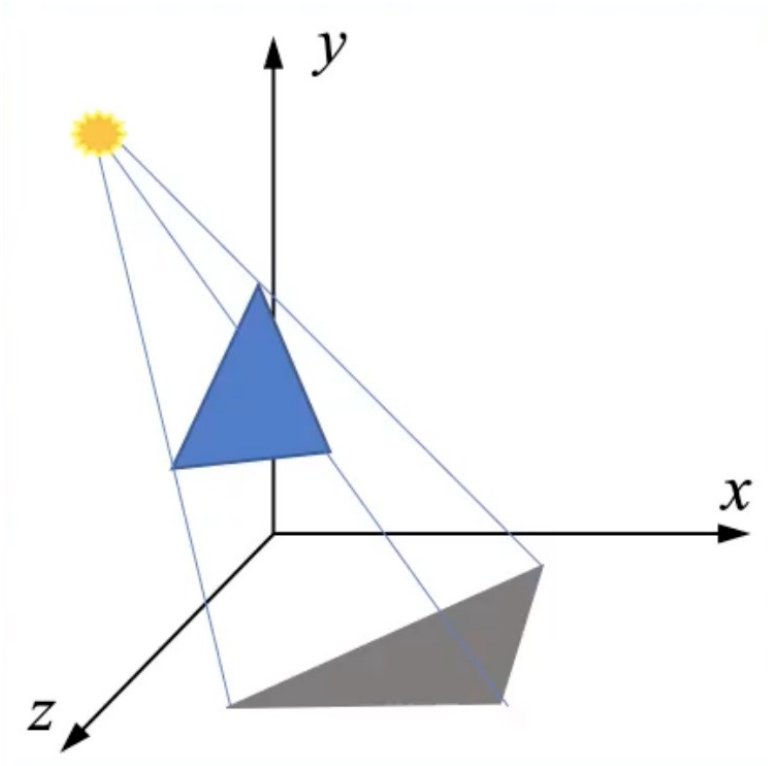


Applications of Planar Homography

- Planar Texture Rendering

Applications of Planar Homography

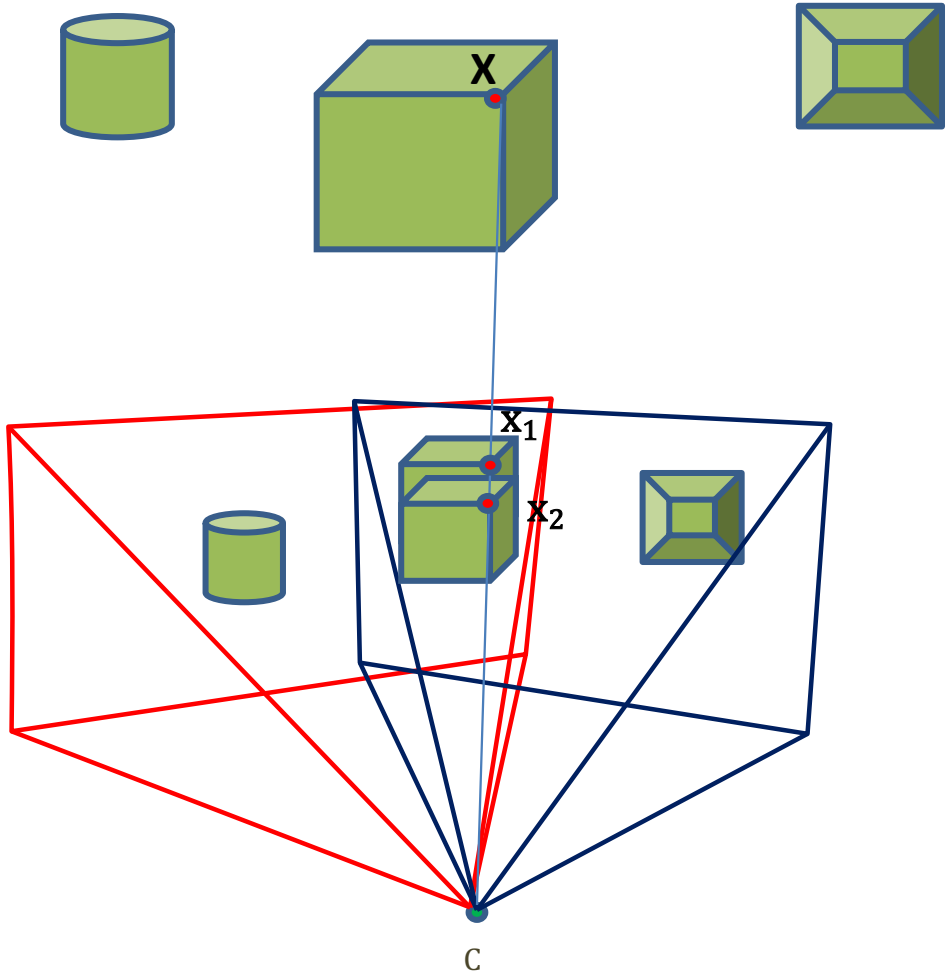
- Rendering Planar Shadows



Applications of Planar Homography

- Estimating relative camera pose for AR

Case 2: Same Camera Center



Arbitrary world

What is the relation between $\mathbf{x}_1$ and $\mathbf{x}_2$?

Case 2: Same Camera Center

- Projection equation for two cameras with same C:

$$\mathbf{x}_1 = \mathbf{K}_1\mathbf{R}_1\begin{bmatrix} \mathbf{I} & \mathbf{-C} \end{bmatrix}\mathbf{X}$$

$$\mathbf{x}_2 = \mathbf{K}_2\mathbf{R}_2\begin{bmatrix} \mathbf{I} & \mathbf{-C} \end{bmatrix}\mathbf{X}$$

$$= \mathbf{K}_2\mathbf{R}_2\left(\mathbf{K}_1\mathbf{R}_1\right)^{-1}\mathbf{K}_1\mathbf{R}_1\begin{bmatrix} \mathbf{I} & \mathbf{-C} \end{bmatrix}\mathbf{X}$$

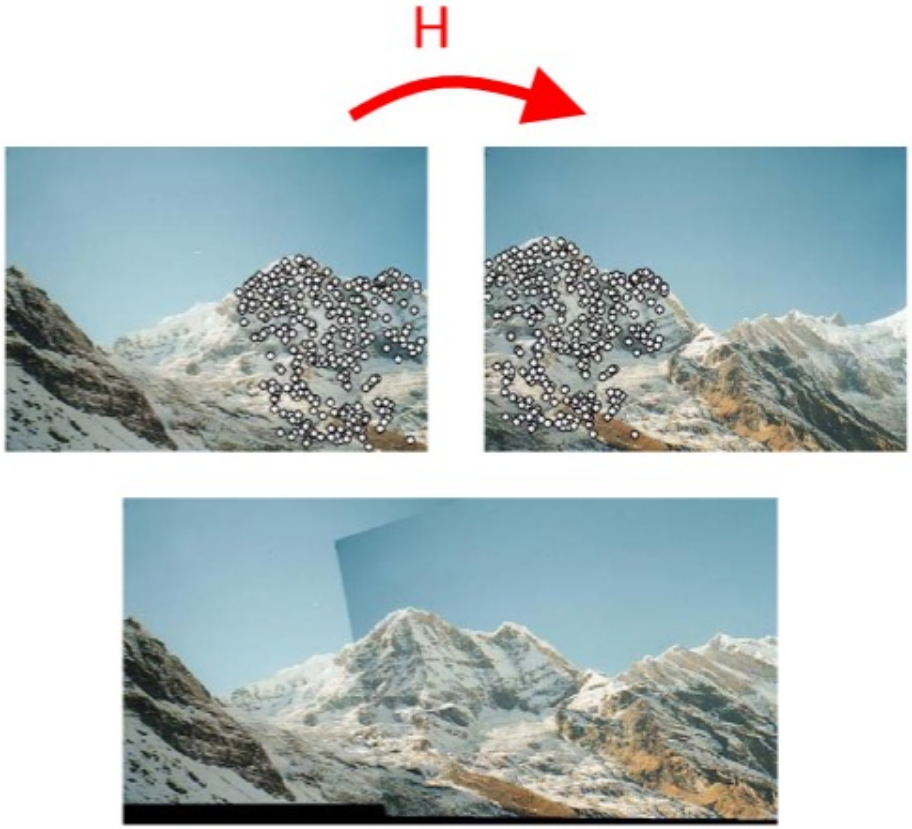
$$= \mathbf{K}_2\mathbf{R}_2\left(\mathbf{K}_1\mathbf{R}_1\right)^{-1}\mathbf{x}_1$$

$$= \mathbf{H}_{21}\mathbf{x}_1$$ where $\mathbf{H}$ is a 3 × 3 non-singular matrix

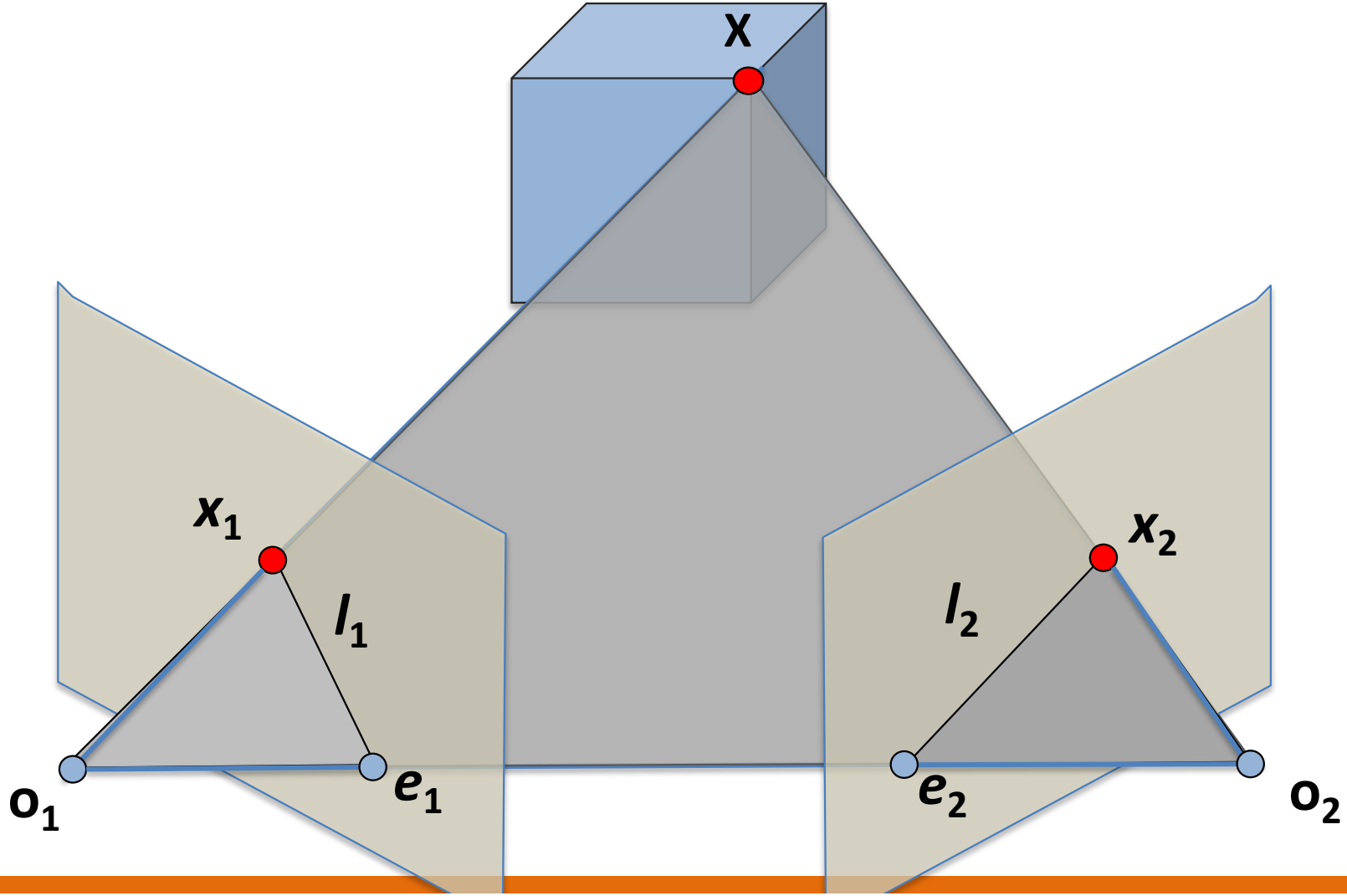
$$\boxed{\mathbf{x}_1 = \mathbf{H}_{12}\mathbf{x}_2; \quad \mathbf{x}_2 = \mathbf{H}_{21}\mathbf{x}_1}$$

Applications of Homography

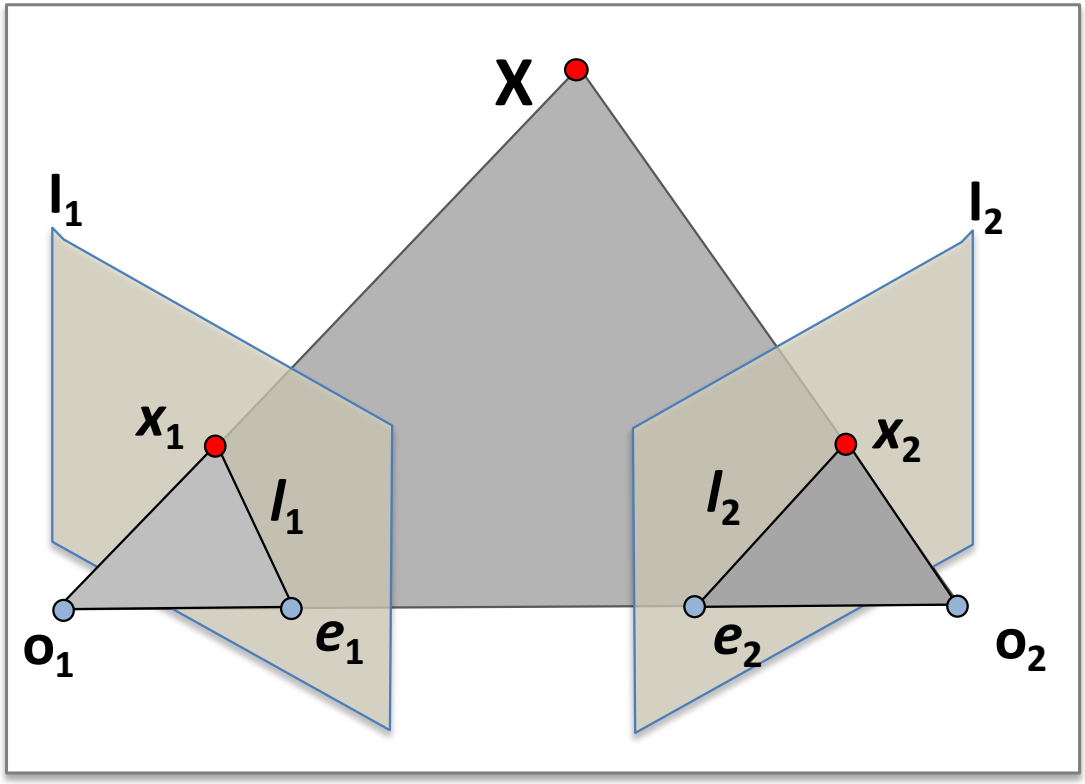
- Image Stitching/Mosaicing, Multi-frame Super-resolution



Case 3:Generic Wrold and Camera

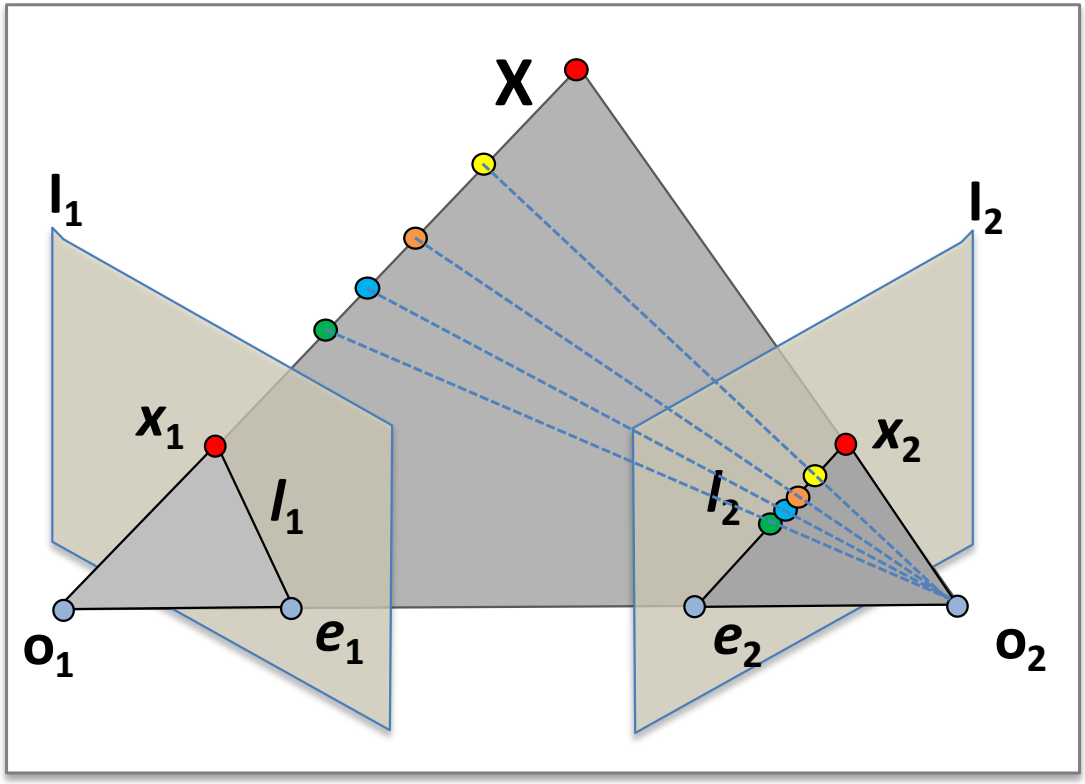


Epipolar Geometry



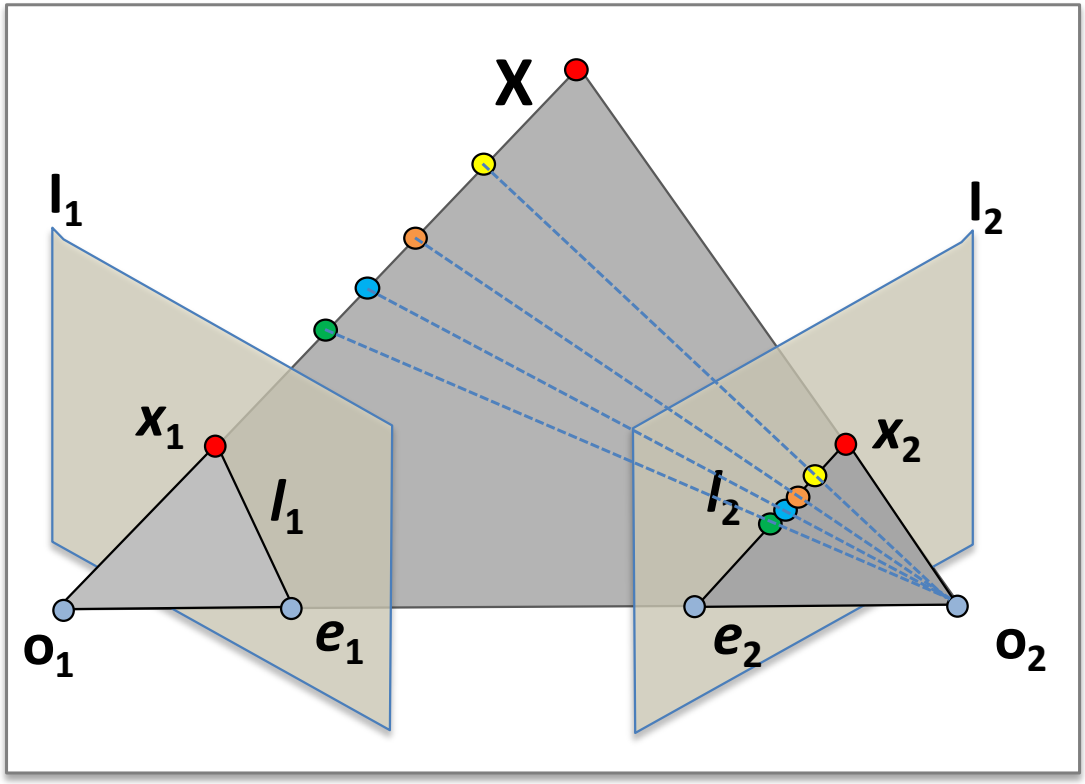
- All world points that map to $\mathbf{x}_1$ in I_1 (pre-image of $\mathbf{x}_1$) map to a line l_2 in I_2, called an epipolar line. (vice-versa)

Epipolar Geometry



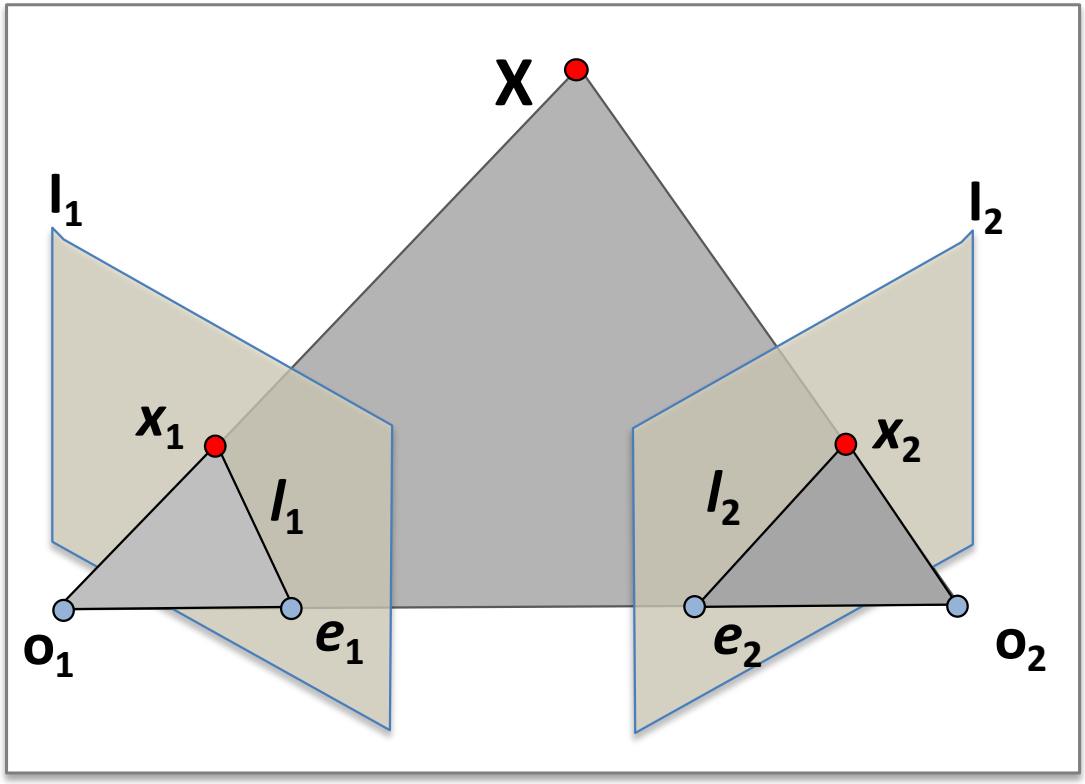
- All world points that map to x_1 in I_1 (pre-image of x_1) map to a line l_2 in I_2, called an epipolar line. (vice-versa)
- The image of o_1 in I_2 (e_2) is an epipole. (vice-versa)

Epipolar Geometry



- The plane containing these is called the epipolar plane.
- These result in a set of constraints, which are referred to as the *epipolar constraints* and the resulting geometry is called the *epipolar geometry.*

Epipolar Constraint: Essential Matrix



$$\lambda_1 \mathbf{x}_1 = \mathbf{X}$$

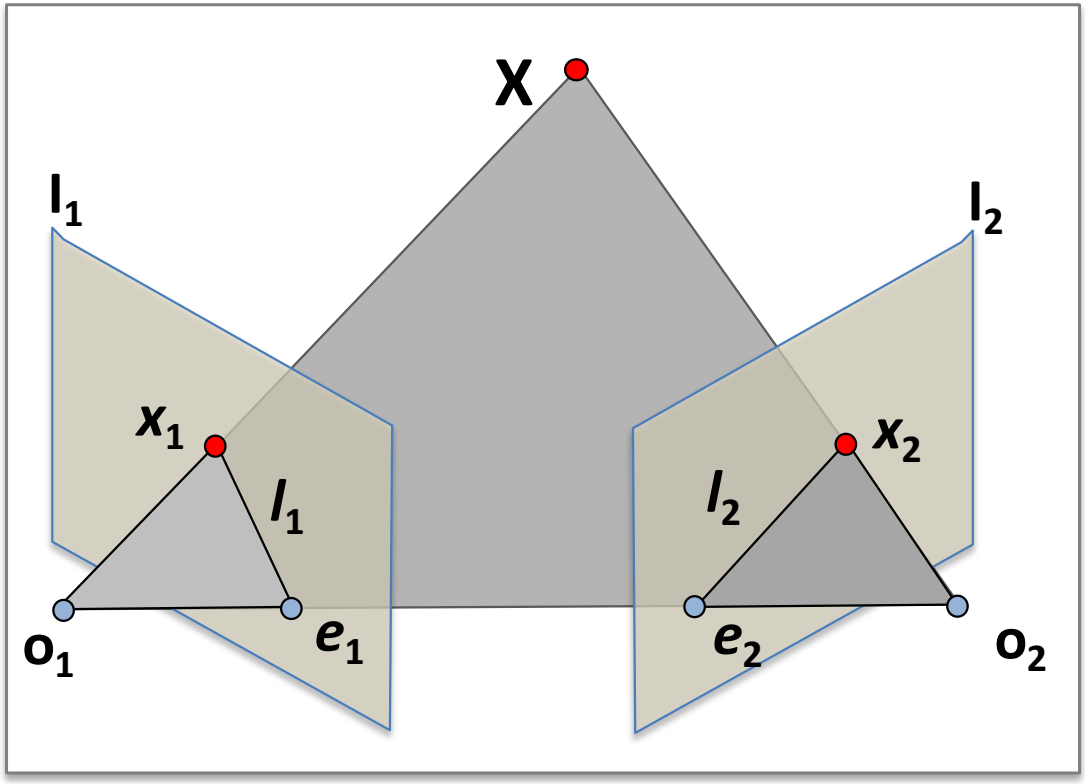
$$\lambda_2 \mathbf{x}_2 = \mathbf{R}\mathbf{X} + \mathbf{T}$$

$$= \mathbf{R}\left(\lambda_1 \mathbf{x}_1\right) + \mathbf{T}$$

Cross product of any two vectors: AxB can be written as the matrix product ÂB, where:

$$\hat{\mathbf{A}} = \begin{bmatrix} 0 & -A_z & A_y \\ A_z & 0 & -A_x \\ A_y & A_x & 0 \end{bmatrix}$$

Epipolar Constraint: Essential Matrix



$$\lambda_1 \mathbf{x}_1 = \mathbf{X}$$

$$\lambda_2 \mathbf{x}_2 = \mathbf{R}\mathbf{X} + \mathbf{T}$$

$$= \mathbf{R}\left(\lambda_1 \mathbf{x}_1\right) + \mathbf{T}$$

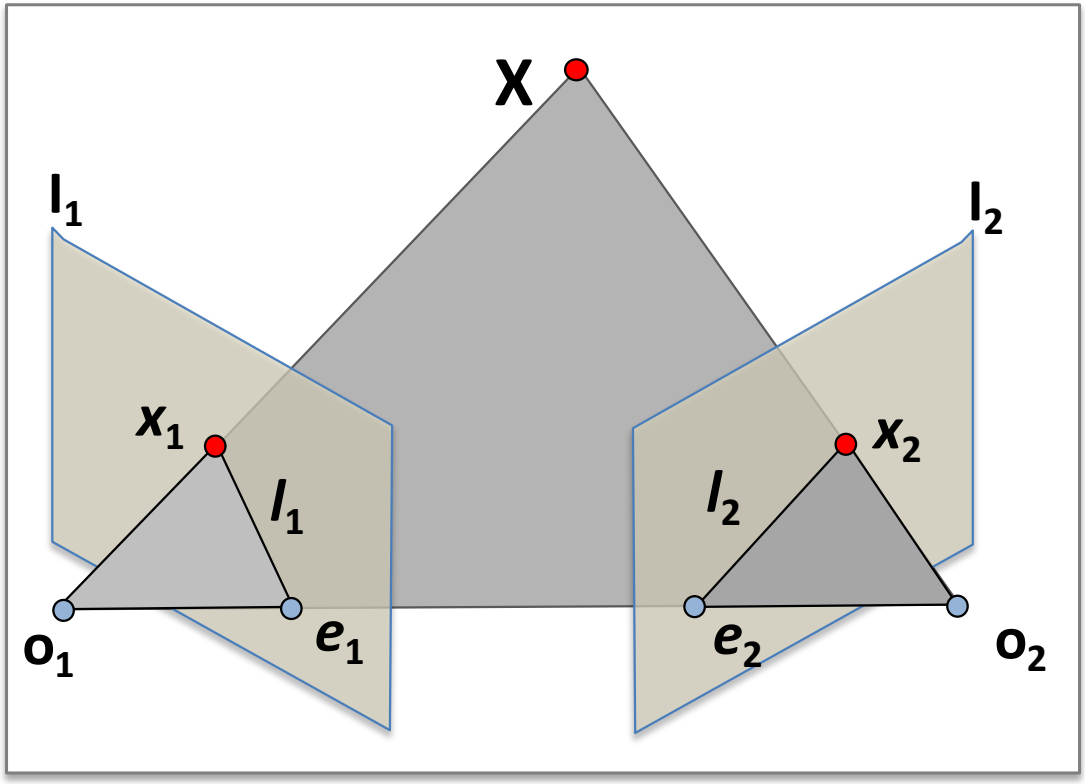
$$\hat{\mathbf{T}}\lambda_2 \mathbf{x}_2 = \hat{\mathbf{T}}\mathbf{R}\lambda_1 \mathbf{x}_1 + 0$$

$$\lambda_2 \mathbf{x}_2^T \hat{\mathbf{T}}\mathbf{x}_2 = \lambda_1 \mathbf{x}_2^T \hat{\mathbf{T}}\mathbf{R}\mathbf{x}_1$$

$$\mathbf{x}_2^T \hat{\mathbf{T}}\mathbf{R}\mathbf{x}_1 = 0$$

$$\mathbf{x}_2^T \mathbf{E}\mathbf{x}_1 = 0 \quad \text{or} \quad \mathbf{x}_1^T \mathbf{E}\mathbf{x}_2 = 0$$

Epipolar Constraint: Fundamental Matrix



$$\mathbf{x}_1 = \mathbf{K}_1 \mathbf{X}$$

$$\mathbf{x}_2 = \mathbf{K}_2 \mathbf{X}$$

$$\mathbf{x}_2^T \mathbf{K}_2^{-T} \hat{\mathbf{T}} \mathbf{R} \mathbf{K}_1^{-1} \mathbf{x}_1 = 0$$

$$\mathbf{x}_1^T F \mathbf{x}_2 = 0$$

Both Essential and Fundamental matrices are 3x3 and are independent of the world point.

Estimate **F** using the 8-point algorithm (SVD)

$$\boldsymbol{x}_m'^{\top}\mathbf{F}\boldsymbol{x}_m = 0$$

$$\begin{bmatrix} x_m' & y_m' & 1 \end{bmatrix} \begin{bmatrix} f_1 & f_2 & f_3 \\ f_4 & f_5 & f_6 \\ f_7 & f_8 & f_9 \end{bmatrix} \begin{bmatrix} x_m \\ y_m \\ 1 \end{bmatrix} = 0$$

ONE correspondence gives you ONE equation

$$\begin{aligned} x_m x_m' f_1 + x_m y_m' f_2 + x_m f_3 + \\ y_m x_m' f_4 + y_m y_m' f_5 + y_m f_6 + \\ x_m' f_7 + y_m' f_8 + f_9 = 0 \end{aligned}$$

$$\begin{bmatrix} x_1 x_1' & x_1 y_1' & x_1 & y_1 x_1' & y_1 y_1' & y_1 & x_1' & y_1' & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_M x_M' & x_M y_M' & x_M & y_M x_M' & y_M y_M' & y_M & x_M' & y_M' & 1 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \\ f_6 \\ f_7 \\ f_8 \\ f_9 \end{bmatrix} = \mathbf{0}$$

https://www.cs.toronto.edu/~lindell/teaching/420/slides/lecture11.pdf

Estimate **F** using the 8-point algorithm (SVD)

How do you solve a homogeneous linear system?

$$\mathbf{A}\boldsymbol{X} = \mathbf{0}$$

Total Least Squares

minimize $\|\mathbf{A}\boldsymbol{x}\|^2$

subject to $\|\boldsymbol{x}\|^2 = 1$

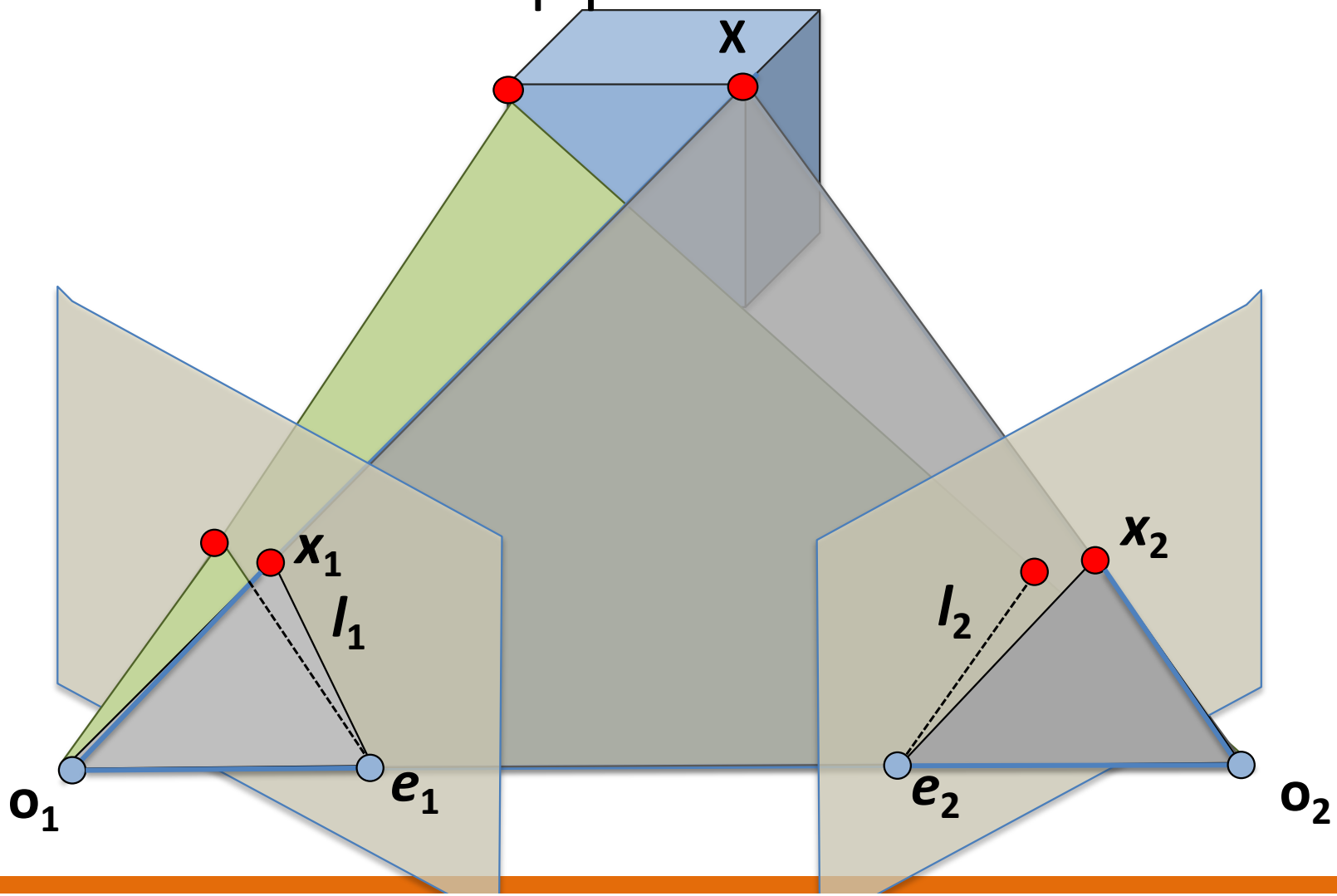
S V D !

0. (Normalize points)
1. Construct the M x 9 matrix **A**
2. Find the SVD of **A**
3. Entries of **F** are the elements of column of **V** corresponding to the least singular value
4. (Enforce rank 2 constraint on F)
5. (Un-normalize F)

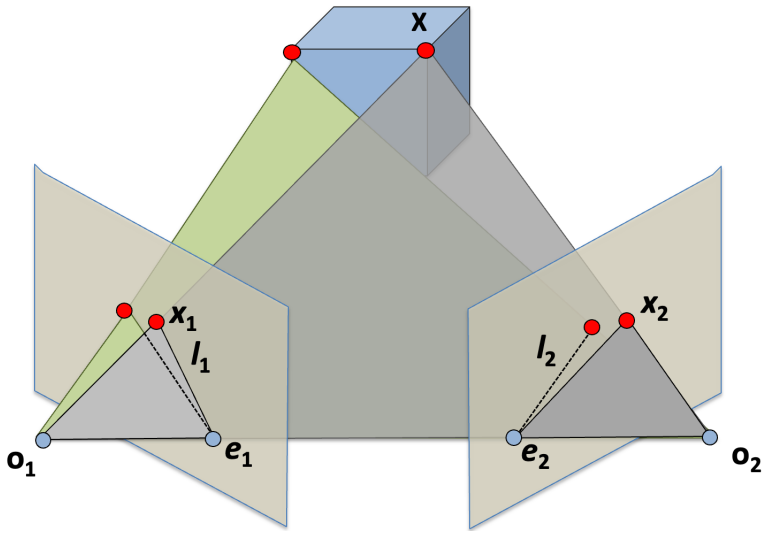
See Hartley-Zisserman for why we do this

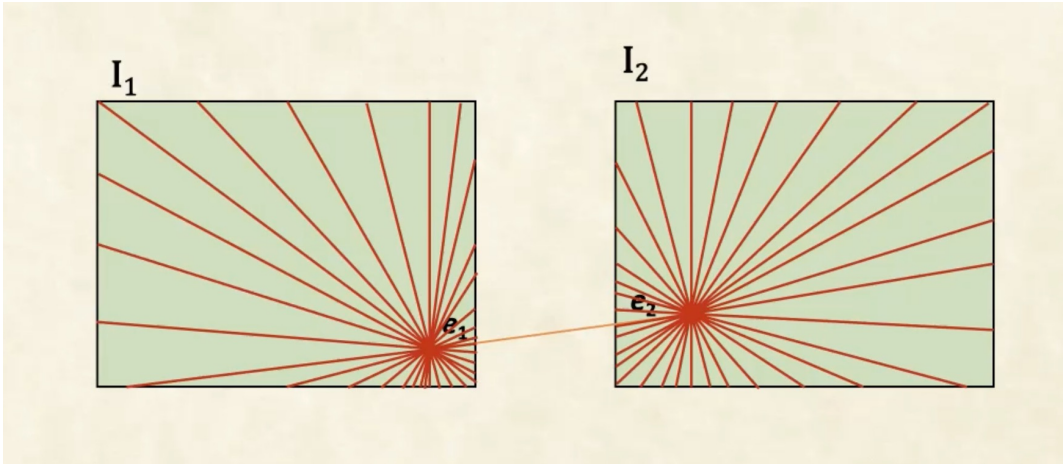
https://www.cs.toronto.edu/~lindell/teaching/420/slides/lecture11.pdf

Epipolar Lines



Epipolar Lines & Rectification





Summary

1. Two views of a planar world are related by a planar Homography. (H: non-singular 3x3 matrix)

1. Two views or an arbitrary world from the same camera position are related by a Homography.

2. Two views of an arbitrary world are constrained by epipolar geometry
    1. Strong Calib.: Essential Matrix: $\boldsymbol{E}$; $x_1{}^T E x_2 = 0$
    2. Weak Calib.: Fundamental Matrix: $\boldsymbol{F}$; $x_1{}^T F x_2 = 0$

Thank You